

SemSafe-3DGS: Semantic Risk-Aware Active Navigation in Uncertain 3D Gaussian Splatting Maps

Amirhossein Mollaei Khass, Athanasios Cosse, Nader Motee

Abstract—Autonomous robots operating in partially observed environments must navigate safely while acquiring observations that improve future planning. Existing safety formulations generally reason primarily about geometry. Consequently, geometrically similar scene elements may induce comparable control responses despite having different semantic consequences. We present a semantic risk aware safe-active perception framework for navigation in attributed 3D Gaussian maps. Semantic attributes modulate an Average Value-at-Risk collision clearance model through class dependent risk weights, allowing safety-critical Gaussian primitives to receive greater influence in the composite barrier. The resulting weighted clearances are aggregated into a control barrier function, while a trajectory-relevant active perception barrier promotes observations that reduce geometric map uncertainty along the robot’s anticipated motion. Both objectives are integrated in a unified CBF-QP that enforces semantic risk-aware collision avoidance as a hard constraint while relaxing information acquisition when it conflicts with safety or task progress. Experiments demonstrate efficient safety constraint, improved navigation through active perception, semantic dependent trajectory adaptation, and real-robot execution under Ackermann dynamics.

I. INTRODUCTION

Autonomous robots navigating unknown environments must make decisions from uncertain maps. An effective policy must jointly determine the robot’s motion and sensing direction to construct a map that supports collision avoidance and task execution. This introduces an inherent conflict, as informative observations often require directing the sensor toward occluded or poorly modeled regions where the corresponding safety margins are least reliable.

3D Gaussian Splatting (3DGS) provides a reliable representation for robotic mapping because it combines explicit spatial primitives with differentiable RGB-D rendering [1]. Recent methods augment Gaussian maps with semantic labels, language features, and task-relevant attributes [2], [3]. However, these attributes are primarily used for recognition and scene understanding rather than safe navigation. Existing 3DGS based navigation methods, however, primarily reason about geometric structure [4]–[7]. Consequently, semantically distinct entities may derive comparable control responses when their geometric configurations are similar.

We propose a semantic risk aware active perception framework for online 3DGS navigation. Semantic attributes modulate the relative collision risk associated with each

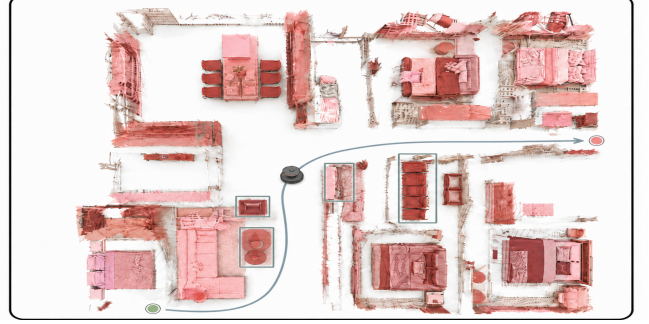


Fig. 1. Semantic 3DGS map with a safe trajectory and nearby semantically labeled objects that influence collision risk.

Gaussian, while the active perception objective focuses on reducing uncertainty near the robot’s future trajectory. A smooth composite safety CBF enforces safety, and perception CBF promotes informative acquisition. The two constraints are integrated in a unified quadratic program that preserves safety as a hard requirement and relaxes perception when conflict occur.

Our contributions are: (i) A semantic risk aware control policy that integrates geometric uncertainty with attributed 3D Gaussian maps. (ii) A smooth composite safety CBF that captures collision risk from all Gaussian obstacles through a single differentiable constraint. (iii) A unified CBF-QP that couples semantic risk-aware safety with trajectory relevant active perception.

II. PROBLEM FORMULATION

We consider a robot navigating toward a goal in an environment with limited prior knowledge. At each control step, the robot receives an onboard RGB-D observation and updates an online Gaussian map. The objective is to preserve safety while moving toward the goal and collect observations that improve the map in regions relevant to future motion. The robot is modeled as the control-affine system

$$\dot{x} = f(x) + g(x)u, \quad (1)$$

where $x \in \mathcal{X} \subset \mathbb{R}^n$ is the state and $u \in \mathcal{U} \subset \mathbb{R}^m$ is the control input. Let $\pi(x) \in \mathbb{R}^3$ and $\eta(x) \in \mathbb{S}^2$ denote the robot position and camera viewing direction, respectively. The environment is represented by an attributed 3DGS map

$$\mathcal{G}_k = \{g_i^k\}_{i=1}^{M_k}, \quad g_i = (\mu_i, R_i, S_i, o_i, c_i, \xi_i), \quad (2)$$

where μ_i , R_i , S_i , o_i , and c_i denote the Gaussian position, orientation, scale, opacity, and appearance. Its covariance is $\Sigma_i = R_i S_i S_i^\top R_i^\top$, and ξ_i denotes its semantic attribute,

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such as a class distribution and a semantic feature with an associated confidence.

Problem Statement. The objective is to learn a control policy that generates safe and informative robot actions by enforcing semantic-risk-aware collision avoidance as a hard constraint while promoting trajectory-relevant information acquisition through a soft perception objective.

III. SEMANTIC-RISK-AWARE SAFE-INFORMATIVE CONTROL

A. Semantic-Risk-Aware Safety Barrier

The safety objective is to prevent the robot from approaching uncertain or semantically hazardous regions represented by the current 3DGS map. Let $d_i(\pi)$ denote the uncertain distance between the robot position and Gaussian g_i . We quantify its lower-tail clearance using Average Value-at-Risk:

$$\text{AV@R}_\varepsilon(d_i^x) := \mathbb{E}[d_i^x \mid d_i^x < \text{VaR}_\varepsilon(d_i^x)] \quad \varepsilon \in (0, 1), \quad (3)$$

where $\text{VaR}_q(d_i)$ is the q -quantile of the distance distribution. The AV@R therefore measures the average clearance within the lower tail of the distance distribution and provides a conservative representation of geometric uncertainty. We define a positive semantic risk weight

$$\kappa(\xi_i) > 0, \quad (4)$$

which maps the semantic attribute and its confidence to the relative collision risk associated with Gaussian g_i . More safety-critical or semantically uncertain objects are assigned smaller values of $\kappa(\xi_i)$, reducing their positive geometric clearance and increasing their influence on the composite safety barrier. The semantic risk aware clearance is then

$$\rho_i^{\text{sem}}(\pi) = \kappa(\xi_i) (\text{AV@R}_\varepsilon(d_i(\pi)) - r_{\text{rob}}). \quad (5)$$

where r_{rob} is the inflated robot radius. A nonnegative value indicates that the lower-tail distance remains outside the collision margins [8].

Safety with respect to the complete Gaussian map requires

$$\min_{g_i \in \mathcal{G}_k} \rho_i^{\text{sem}}(\pi) \geq 0. \quad (6)$$

Because the pointwise minimum is nonsmooth, we approximate it by the composite safety barrier

$$h_s(\pi) = -\frac{1}{\beta_s} \log \left(\sum_{g_i \in \mathcal{G}_k} \exp[-\beta_s \rho_i^{\text{sem}}(\pi)] \right), \quad \beta_s > 0. \quad (7)$$

For M_k Gaussian primitives [6], [9],

$$\min_i \rho_i^{\text{sem}} - \frac{\log M_k}{\beta_s} \leq h_s \leq \min_i \rho_i^{\text{sem}}. \quad (8)$$

Thus, $h_s(\pi) \geq 0$ is a smooth sufficient condition for maintaining nonnegative semantic risk of collision. For the dynamics in (1), the corresponding safety CBF condition is

$$L_f h_s(x) + L_g h_s(x)u \geq -\gamma_s \alpha_s(h_s(x)), \quad (9)$$

where α_s is an extended class- $\mathcal{K}$ function and $\gamma_s > 0$. The soft-min structure also provides smooth attention over the 3DGS map:

$$\nabla_\pi h_s = \sum_{i=1}^{M_k} \omega_i \nabla_\pi \rho_i^{\text{sem}}, \quad \omega_i = \frac{e^{-\beta_s \rho_i^{\text{sem}}}}{\sum_j e^{-\beta_s \rho_j^{\text{sem}}}}. \quad (10)$$

Consequently, the control response is dominated by Gaussian primitives with the smallest semantic-weighted clearance. By assigning smaller positive values of $\kappa(\xi_i)$ to higher-risk semantic classes, such primitives receive greater weight in the composite barrier and can influence the control action earlier than geometrically similar benign objects.

B. Trajectory-Relevant Active-Perception Barrier

Safety alone does not guarantee that the robot acquires observations that improve future planning. We therefore retain a trajectory-relevant geometric active-perception objective that evaluates information gain only within regions associated with the robot's anticipated motion. Let $\mathcal{P}_k = \{\bar{p}_0, \bar{p}_1, \dots, \bar{p}_H\}$, denote a nominal trajectory by u_k^{ref} , where $\bar{p}_0 = \pi(x_k)$. In each waypoint, we define

$$r_\tau = r_{\min} + \beta_1 \exp \left(-\beta_2 \min_{g_i \in \mathcal{G}_k} \rho_i^{\text{sem}}(\bar{p}_\tau) \right), \quad (11)$$

The trajectory relevant map region is

$$\Omega_k = \bigcup_{\tau=0}^H \mathcal{B}(\bar{p}_\tau, r_\tau), \quad \mathcal{G}_k|_\Omega = \{g_i \in \mathcal{G}_k \mid \mu_i \in \Omega_k\}. \quad (12)$$

This restricts information evaluation to map regions likely to affect future navigation.

Let w_Ω denote the parameters of $\mathcal{G}_k|_\Omega$ and Y be the RGB-D observation predicted from sensing pose (π, η) , the observed Fisher information is approximated by

$$\mathcal{H}((\pi, \eta)) = -\nabla_{w_\Omega}^2 \log p(Y \mid (\pi, \eta), w_\Omega) \big|_{w_\Omega = w_\Omega^*}. \quad (13)$$

Let $\mathcal{H}_{\text{prior}} > 0$ denote the regularized information accumulated from previous observations. We define the trajectory relevant information utility as

$$\mathcal{I}((\pi, \eta)) = \text{tr} \left(\mathcal{H}((\pi, \eta)) \mathcal{H}_{\text{prior}}^{-1} \right). \quad (14)$$

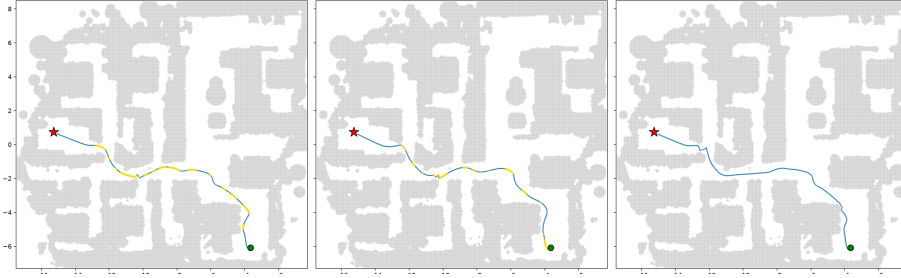
The information score increases when the candidate sensing pose is expected to reduce uncertainty in the Gaussian parameters associated with the robot's nominal future trajectory [8], [10]. The active perception barrier is therefore defined as

$$h_p(\pi, \eta) = \mathcal{I}(\pi, \eta) - \mathcal{I}_c, \quad (15)$$

where $\mathcal{I}_c > 0$ is a desired information threshold. Its CBF condition is

$$L_f h_p(x) + L_g h_p(x)u \geq -\gamma_p \alpha_p(h_p(x)) - \delta, \quad \delta \geq 0. \quad (16)$$

The slack variable allows information acquisition to be relaxed when it conflicts with collision avoidance, control limits, or nominal task progress.



	Ours	CAAP	No perc.
Goal time [s]	74	82	103
Path length [m]	16.6	16.9	16.4
Mean speed [m/s]	0.212	0.192	0.143
Mean h_s	+0.724	+0.680	+0.561
Mean h_p	+0.005	-0.275	—

Fig. 2. Safe-active navigation comparison in InteriorGS: proposed method (left), CAAP [6] (middle), and safety-only navigation (right). Yellow trajectory segments indicate active perception. Our method achieves the shortest goal time and highest mean speed while maintaining positive mean safety and perception barriers.

Method	Scene	Safety [%] ↑	Time [ms] ↓	Min. Dist. [m] ↑	Control Dev. ↓
SAFER-Splat	Stonehenge	100	30.9	0.086	0.069
	Statues	99	45.0	0.141	0.028
	Flightgate	99	60.3	0.086	0.062
	Adirondacks	100	186.0	0.079	0.046
Ours $c = 0.90$	Stonehenge	100	2.64	0.120	0.105
	Statues	99	2.68	0.169	0.104
	Flightgate	98	2.72	0.150	0.098
	Adirondacks	98	2.79	0.111	0.098
Ours $c = 0.75$	Stonehenge	100	1.63	0.106	0.070
	Statues	99	1.49	0.158	0.024
	Flightgate	99	1.50	0.143	0.064
	Adirondacks	99	1.84	0.075	0.044

TABLE I

SAFETY-BARRIER COMPARISON ACROSS FOUR 3DGS SCENES.

C. Unified Semantic Safe-Active CBF-QP

The next safe-informative control policy is obtained from

$$\begin{aligned}
 (u_k^*, \delta_k^*) = \arg \min_{u, \delta} \quad & \frac{1}{2} \|u - u_k^{\text{ref}}\|_2^2 + \frac{\lambda_\delta}{2} \delta^2 \\
 \text{s.t.} \quad & L_f h_s + L_g h_s u \geq -\gamma_s \alpha_s(h_s), \\
 & L_f h_p + L_g h_p u \geq -\gamma_p \alpha_p(h_p) - \delta, \\
 & u_{\min} \preceq u \preceq u_{\max}, \quad \delta \geq 0,
 \end{aligned} \tag{17}$$

where $\lambda_\delta > 0$ controls the cost of relaxing the perception objective. The first constraint enforces semantic-risk-aware collision avoidance and is never relaxed. The second constraint promotes geometric information acquisition near the robot’s anticipated trajectory but permits controlled violation through δ . The objective minimally modifies the nominal task policy while penalizing the loss of perception performance. The resulting controller therefore prioritizes safety around geometrically uncertain or semantically hazardous regions while continuing to improve the map whenever informative sensing is compatible with safe task execution.

IV. EXPERIMENTS

We evaluate the proposed framework through different complementary studies: (i) computational and performance of the composite safety barrier, (ii) the effect of trajectory-relevant active perception on closed-loop navigation, and (iii) real-robot deployment under Ackermann dynamics.

A. Safety Barrier Evaluation

We first isolate the safety component using double-integrator dynamics and compare it with SAFER-Splat [5],

which constructs collision constraints from the distance between the robot and individual Gaussian ellipsoids. For a controlled comparison, all methods use the same PD nominal controller, dynamics, and 3DGS maps trained with Splatfacto [11]: Stonehenge (100K Gaussians), Statues (200K), Flightgate (300K), and Adirondacks (500K). As reported in Table I, SAFER-Splat achieves an average safety rate of 99.5%, compared with 98.75% and 99.25% for our barrier with $c = 1 - \epsilon \in \{0.75, 0.90\}$, respectively. A more conservative risk estimate enlarges the effective obstacle region and can slightly reduce the success rate in cluttered scenes by restricting admissible motions. In contrast, the average online computation decreases from 80.55 ms to 2.71 ms and 1.62 ms, corresponding to approximately $30\times$ and $50\times$ speedups. The proposed barrier also maintains larger average minimum clearances for both settings, demonstrating that the smooth map-level aggregation substantially reduces computation without compromising collision avoidance.

B. Trajectory-Relevant Active Perception

We next compare the complete controller with CAAP [6] and a safety-only baseline in the InteriorGS [12] environment using unicycle dynamics. The safety-only controller removes the perception constraint and follow the nominal trajectory safety without any information seeking, while CAAP promotes alignment with the expected information gain ascend direction. All methods use identical start–goal states, control limits, online RGB-D map updates, and A* replanning as the initially partial Gaussian map evolves. As shown in Fig. 2, our method reaches the goal in 74 s, compared with 82 s for CAAP and 103 s without active perception, despite comparable path lengths (16.6, 16.9, and 16.4 m). The improvement is instead reflected in a higher mean velocity (0.212 m/s) while maintaining the largest mean safety barrier ($\bar{h}_s = 0.724$). Moreover, the proposed trajectory-conditioned objective maintains a positive mean perception barrier ($\bar{h}_p = 0.005$), whereas CAAP yields $\bar{h}_p = -0.275$. These results indicate that directing sensing toward uncertainty relevant to anticipated motion reduces conservative behavior without sacrificing safety.

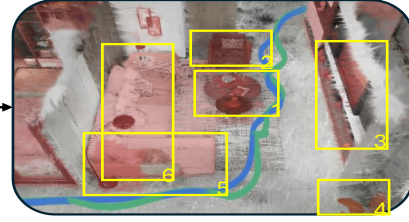
C. Real-Robot Ackermann Validation

Finally, we deploy the controller on a physical Ackermann steered mobile robot operating in an indoor 3DGS map.



Controller	Goal	Time [s] ↓	Path [m] ↓	Speed [cm/s] ↑	Min h_s [m] ↑	Avg h_s [m] ↑	High-Risk Clearance [cm] ↑
Geometric	✓	59.8	20.95	35.1	0.026	0.553	14.8
Semantic	✓	86.2	20.97	24.4	0.033	0.435	36.7

(b) Zoomed semantic region



1: Vase
2: Couch
3: TV
4: Plant
5: Lamp
6: Couch

Fig. 3. Semantic-risk-aware navigation in an attributed 3DGS map. (a) Executed trajectories with geometric safety + perception (blue) and semantic-aware safety + perception (green). (b) Enlarged region showing semantic object annotations used to modulate class-dependent collision-risk weights.

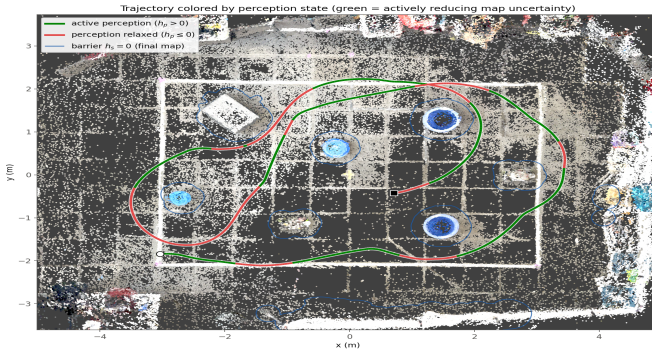


Fig. 4. Real-robot Ackermann navigation in an indoor 3DGS map. The executed trajectory is colored by the perception state: green denotes active perception ($h_p > 0$) and red denotes perception relaxation ($h_p \leq 0$). Blue contours indicate the safety boundary $h_s = 0$. The controller regulates motion around unsafe regions while progressing from the start to the goal.

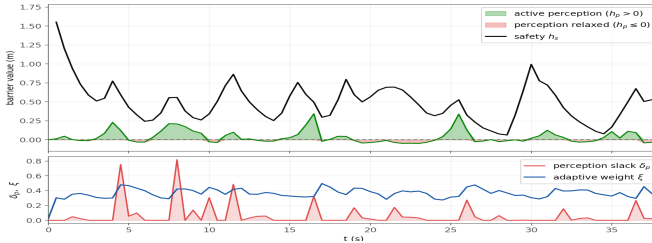


Fig. 5. Barrier and perception-relaxation signals during the real-robot experiment. The safety barrier h_s remains positive throughout the run, while h_p , the perception slack δ_p , and adaptive weight ξ characterize the online safety-perception interaction.

Unlike the unicycle simulation, the platform is subject to nonholonomic steering, bounded acceleration, and limited braking authority; the safety condition is therefore implemented through a higher order barrier while retaining the soft trajectory relevant perception constraint. Fig. 4 shows that the controller departs from the nominal motion near safety-critical regions while continuing toward the goal. Over the 38 s run, the safety barrier remains strictly positive with a minimum value of $h_s = 0.046$ m; Fig. 5 further shows the interaction between safety enforcement and perception relaxation during execution. The hardware results demonstrate that the proposed map-level controller can be executed online

under nonholonomic vehicle dynamics while maintaining the prescribed safety condition.

D. Semantic aware safe active perception

To incorporate semantics into safety-critical navigation, each Gaussian is augmented with a semantic attribute ξ_i . Using InteriorGS [12], we associate each primitive with the annotated object instance containing its center μ_i and assign the corresponding semantic class; unmatched primitives are labeled *unknown*. Each semantic class c is mapped to a positive risk-scaling factor $\kappa(c)$, with smaller values assigned to classes requiring more conservative treatment. The resulting semantic weighting modifies the relative contribution of each Gaussian to the composite safety barrier in (7). More generally, semantic attributes may be transferred from 3D annotations or lifted from image-level predictions [2], [13], [14]; for datasets such as Gibson [15], they can be aggregated from pixel-wise semantic observations across views. Thus, the controller is agnostic to the semantic estimation pipeline and can incorporate semantic or VLM derived attributes as class dependent collision-risk weights. [16]–[18].

We evaluate the effect of semantic collision risk by comparing the proposed semantic-aware controller with the same safe-active perception policy using geometry-only safety. As shown in Fig. 3, semantic attributes modify the relative risk weighting of nearby Gaussian primitives and consequently alter the executed trajectory around semantically safety-critical objects. Quantitative differences in path length, goal time, and the mean and minimum safety barriers are reported in the figure.

V. CONCLUSION

We presented a semantic-risk-aware safe-active perception framework for navigation in attributed 3D Gaussian maps. Semantic attributes scale the risk-adjusted clearance of individual Gaussian primitives, allowing the composite safety CBF to prioritize obstacles according to both geometric uncertainty and semantic collision consequence. Experiments demonstrate navigation efficiency through active perception, semantic-dependent changes in the executed trajectory.

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